

Boolean Algebra

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Chapter Summary

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- 2 Boolean Functions
- 3 Representing Boolean Functions
- 4 Logic Gates
- 5 Minimization of Circuits

Introduction

- Boolean algebra traces its origins to an 1854 book by the mathematician George Boole.
- Boolean algebra is a division of mathematics that deals with operations on logical values and incorporates binary variables.
- Boolean algebra can be viewed as a foundation for designing and analyzing digital systems!
- Electronic and optical switches can be studied using this set and the rules of Boolean algebra.
- Boolean algebra provides the operations and rules for working with the set $\{0, 1\}$.

Operations

Operations:

- Three basic operations in Boolean algebra

- the Boolean sum, denoted by $+$ or by *OR*.

$$1 + 1 = 1, \quad 1 + 0 = 1, \quad 0 + 1 = 1, \quad 0 + 0 = 0$$

- the Boolean product, denoted by $\cdot$ or by *AND*.

$$1 \cdot 1 = 1, \quad 1 \cdot 0 = 0, \quad 0 \cdot 1 = 0, \quad 0 \cdot 0 = 0$$

- the complement of an element, denoted with a bar $\bar{x}$.

$$\bar{0} = 1$$

$$\bar{1} = 0$$

Precedence

Precedence rules:

1 -> all complements

2 -> all products

3 -> all sums

Example:

Find the value of $1 \cdot 0 + \overline{(0 + 1)}$.

Solution: Using the definitions of complementation, the Boolean sum, and the Boolean product, it follows that

$$1 \cdot 0 + \overline{(0 + 1)} = 0 + \overline{1} = 0 + 0 = 0$$

Boolean Algebra Vs Logic

Complement	$\neg$
Boolean sum	$\vee$
Boolean product	$\wedge$
0	F (false)
1	T (true)

Example: Translate $1 \cdot 0 + (0 + 1)$, into a logical equivalence.

Solution: $1 \cdot 0 + (0 + 1) \implies (T \wedge F) \vee \neg(T \vee F) \equiv F$.

Example: Translate the logical equivalence $(T \wedge T) \vee \neg F \equiv T$ into an identity in Boolean algebra.

Solution: $(T \wedge T) \vee \neg F \equiv T \implies (1 \cdot 1) + \bar{0} = 1$.

Boolean functions

Boolean Expressions and Boolean Functions

- Let $B = \{0, 1\}$.
- $B_n = \{(x_1, x_2, \dots, x_n) | x_i \in B \text{ for } 1 \leq i \leq n\}$ is the set of all possible n -tuples of 0s and 1s.
- Boolean variable: assume values only from B , that is, 0 and 1
- Boolean function of degree n : $F : B_n \rightarrow B$

Example: $F(x, y) = x\bar{y}$, a Boolean function of degree 2 with $F(1, 1) = 0$, $F(1, 0) = 1$, $F(0, 1) = 0$, and $F(0, 0) = 0$.

x	y	$F(x, y)$
1	1	0
1	0	1
0	1	0
0	0	0

Boolean Functions

- Boolean functions can be represented using expressions made up of variables and Boolean operations.
- The Boolean expressions in the variables $x_1, x_2, \dots, x_n$ are defined recursively as follows.
 - $0, 1, x_1, x_2, \dots, x_n$ are Boolean expressions;
 - if E_1 and E_2 are Boolean expressions, then $\overline{E_1}$, $(E_1 \cdot E_2)$, and $(E_1 + E_2)$ are Boolean expressions.
- Boolean expression represents a Boolean function.
- The values of this function are obtained by substituting 0 and 1 for the variables in the expression.

Boolean Functions

Example: Find the values of the Boolean function represented by $F(x, y, z) = xy + \bar{z}$.

Solution:

x	y	z	xy	$\bar{z}$	$F(x, y, z) = xy + \bar{z}$
1	1	1	1	0	1
1	1	0	1	1	1
1	0	1	0	0	0
1	0	0	0	1	1
0	1	1	0	0	0
0	1	0	0	1	1
0	0	1	0	0	0
0	0	0	0	1	1

Boolean Functions

Let F and G be boolean functions of variables n .

Equal: $F(b_1, b_2, \dots, b_n) = G(b_1, b_2, \dots, b_n)$ whenever $b_1, b_2, \dots, b_n$ belong to B .

Equivalent: Represent the same function.

Example: xy , $xy + 0$, and $xy \cdot 1$ are equivalent.

Complement: $\overline{F}$, where $\overline{F}(x_1, \dots, x_n) = \overline{F(x_1, \dots, x_n)}$.

Boolean sum: $F + G$

$$(F + G)(x_1, \dots, x_n) = F(x_1, \dots, x_n) + G(x_1, \dots, x_n)$$

Boolean product: $(F \cdot G)$

$$(F \cdot G)(x_1, \dots, x_n) = F(x_1, \dots, x_n) \cdot G(x_1, \dots, x_n)$$

Identities of Boolean Algebra

- Useful in simplifying the design of circuits.
- Can be proved using a truth table.

Identity	Name
$\overline{\overline{x}} = x$	double complement
$x + x = x$ $x \cdot x = x$	Idempotent
$x + 0 = x$ $x \cdot 1 = x$	Identity
$x + 1 = 1$ $x \cdot 0 = 0$	Domination
$x + y = y + x$ $xy = yx$	Commutative

Identity	Name
$x + (y + z) = (x + y) + z$ $x(yz) = (xy)z$	Associative
$x + yz = (x + y)(x + z)$ $x(y + z) = xy + xz$	Distributive
$\overline{(xy)} = \overline{x} + \overline{y}$ $\overline{(x + y)} = \overline{x}\overline{y}$	De Morgan's
$x + xy = x$ $x(x + y) = x$	Absorption
$x + \overline{x} = 1$	Unit property
$x\overline{x} = 0$	Zero property

Proof of Identities: Truth table

Can be proved using a table.

Example: Proof of distributive law $x(y + z) = xy + xz$

Solution:

x	y	z	$y + z$	xy	yz	$x(y + z)$	$xy + yz$
1	1	1	1	1	1	1	1
1	1	0	1	1	0	1	1
1	0	1	1	0	1	1	1
1	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	0	1	1	0	0	0	0
0	0	0	0	0	0	0	0

The identity holds because the last two columns agree.

Logical Equivalence Expression:

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

Proof of Identities: Using identities

Identities in Boolean algebra can be used to prove further identities.

Example: Prove the absorption law $x(x + y) = x$ using the other identities of Boolean algebra.

Solution:

$$\begin{aligned}x(x + y) &= (x + 0)(x + y) \\&= x + 0 \cdot y \\&= x + y \cdot 0 \\&= x + 0 \\&= x\end{aligned}$$

Identity law for sum

Distributive law of sum over product

Commutative law for product

Domination law for product

Identity law for sum

Boolean Functions: Duality

Dual of a Boolean expression:

- Interchanging Boolean sums and Boolean products
- Interchanging 0's and 1's

Example: Find the duals of $x(y + 0)$ and $\bar{x} \cdot 1 + (\bar{y} + z)$.

Solution: The duals are $x + (y \cdot 1)$ and $(\bar{x} + 0)(\bar{y}z)$

Boolean Algebra: Formal Denition

The Abstract Definition of a Boolean Algebra:

Definition

A Boolean algebra is a set B with two binary operations $\vee$ and $\wedge$, elements 0 and 1 , and a unary operation $\bar{}$ such that these properties hold for all x , y , and z in B .

$x \vee 0 = x$ $x \wedge 1 = x$	Identity laws
$x \vee \bar{x} = 1$ $x \wedge \bar{x} = 0$	Complement

$x \vee y = y \vee x$ $x \wedge y = y \wedge x$	Commutative
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$(x \vee y) \vee z = x \vee (y \vee z)$ $(x \wedge y) \wedge z = x \wedge (y \wedge z)$	Associative
$x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$ $x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$	Distributive

Sum-of-Products Expansions

Also known as **disjunctive normal form (dnf)**

x	y	z	F	G
1	1	1	0	0
1	1	0	0	1
1	0	1	1	0
1	0	0	0	0
0	1	1	0	0
0	1	0	0	1
0	0	1	0	0
0	0	0	0	0

Example: Find Boolean expressions that represent the functions $F(x, y, z)$ and $G(x, y, z)$, which are given in the table.

Solution: For F , an expression that has the value 1 when $x = z = 1$ and $y = 0$, and the value 0 otherwise. The product, $x\bar{y}z$, has the value 1 if and only if $x = \bar{y} = z = 1$, which holds if and only if $x = z = 1$ and $y = 0$.

For G , we need an expression that equals 1 when $x = y = 1$ and $z = 0$, or $x = z = 0$ and $y = 1$. The Boolean sum of these two products, $xy\bar{z} + \bar{x}\bar{y}\bar{z}$, represents G .

Literal and Minterm

Literal: A literal is either a Boolean variable or its complement.

Minterm: A minterm of the Boolean variables $x_1, x_2, \dots, x_n$ is a Boolean product $y_1 y_2 \dots y_n$, where $y_i = x_i$ or $y_i = \overline{x_i}$. Hence, a minterm is a product of n literals, with one literal for each variable.

A minterm has the value 1 for one and only one combination of values of its variables. More precisely, the minterm $y_1 y_2 \dots y_n$ is 1 if and only if each y_i is 1, and this occurs if and only if $x_i = 1$ when $y_i = x_i$ and $x_i = 0$ when $y_i = \overline{x_i}$.

Example: Find a minterm that equals 1 if $x_1 = x_3 = 0$ and $x_2 = x_4 = x_5 = 1$, and equals 0 otherwise.

Solution: The minterm $\overline{x_1} x_2 \overline{x_3} x_4 x_5$ has the correct set of values.

sum-of-products example

Sum-of-products expansion (dnf): sum of minterms

Example: Find the sum-of-products expansion for the function

$$F(x, y, z) = (x + y)\bar{z}.$$

Solution: One way: Using Boolean identities to expand the product and simplify.

$$F(x, y, z) = (x + y)\bar{z}$$

$$= x\bar{z} + y\bar{z}$$

Distributive law

$$= x1\bar{z} + 1y\bar{z}$$

Identity law

$$= x(y + \bar{y})\bar{z} + (x + \bar{x})y\bar{z}$$

Unit property

$$= xy\bar{z} + x\bar{y}\bar{z} + xy\bar{z} + \bar{x}y\bar{z}$$

Distributive law

$$= xy\bar{z} + x\bar{y}\bar{z} + \bar{x}y\bar{z}$$

Idempotent law

sum-of-products example contd...

Example: Find the sum-of-products expansion for the function $F(x, y, z) = (x + y)\bar{z}$.

Solution: Second way: Using truth table. The sum-of-products expansion of F is the Boolean sum of minterms corresponding to the rows of this table that give the value 1 for the function.

x	y	z	$x + y$	$\bar{z}$	$(x + y)\bar{z}$
1	1	1	1	0	0
1	1	0	1	1	1
1	0	1	1	0	0
1	0	0	1	1	1
0	1	1	1	0	0
0	1	0	1	1	1
0	0	1	0	0	0
0	0	0	0	1	0

$$xy\bar{z} + x\bar{y}\bar{z} + \bar{x}y\bar{z}$$

Maxterm

Maxterm: A maxterm of the Boolean variables $x_1, x_2, \dots, x_n$ is a Boolean sum $y_1 + y_2 + \dots + y_n$, where $y_i = x_i$ or $y_i = \overline{x_i}$. Hence, a maxterm is a sum of n literals, with one literal for each variable.

A maxterm has the value 0 for one and only one combination of values of its variables. More precisely, the maxterm $y_1 + y_2 + \dots + y_n$ is 0 if and only if each y_i is 0, and this occurs if and only if $x_i = 0$ when $y_i = x_i$ and $x_i = 1$ when $y_i = \overline{x_i}$.

Example: Find a maxterm that equals 0 if $x_1 = x_3 = 1$ and $x_2 = x_4 = x_5 = 0$, and equals 1 otherwise.

Solution: The maxterm $\overline{x_1} + x_2 + \overline{x_3} + x_4 + x_5$ has the correct set of values.

Product-of-sums expansions

Also known as **conjunctive normal form (cnf)**

Dual of sums-of-products

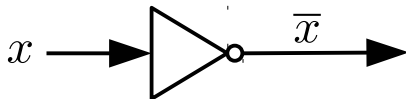
Products-of-sums expansion (dnf): Product of maxterms

Logic Gates

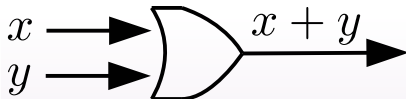
- Boolean algebra is used to model circuitry of electronic devices.
- Each input or output is in the set $\{0, 1\}$.
- Circuit can be designed using the rules of Boolean algebra.
- The basic elements of circuits are called **gates**.
- Each type of gate implements a Boolean operation.
- These circuits have no memory capabilities. Such circuits are called **combinational circuits** or **gating networks**.

Logic Gates: Basic Gates

Inverter/NOT gate



OR gate

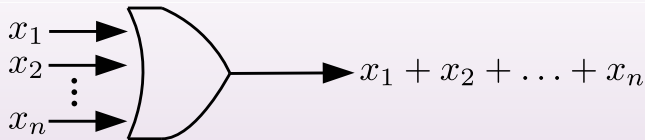
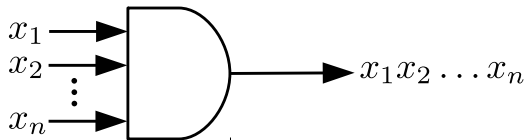


AND gate



Logic gates: multiple inputs

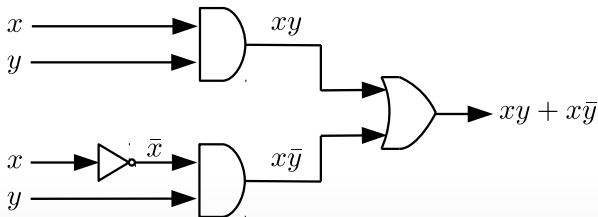
Multiple inputs to the AND and OR gates.



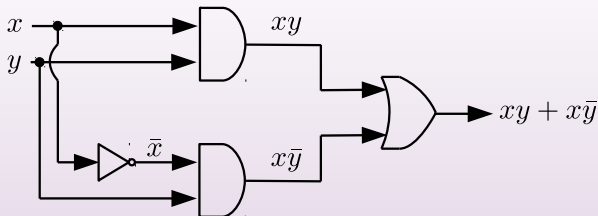
Logic gates: combinations of gates

Combinational circuits can be constructed using a combination of inverters, OR gates, and AND gates.

Separate input
for each gate



Sharing the input

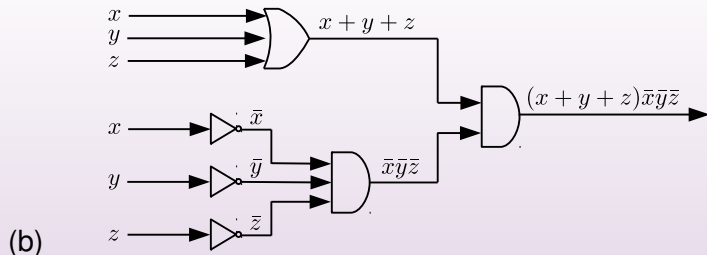
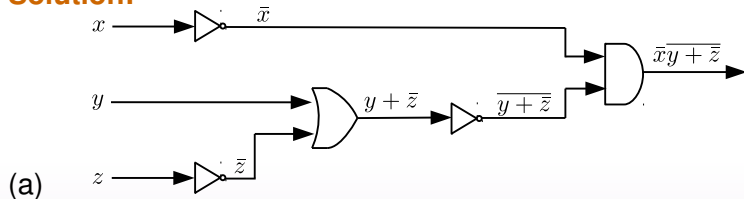


Combinations of gates: example

Example: Construct circuits that produce the following outputs:

(a) $\bar{x}(y + \bar{z})$, and (b) $(x + y + z)(\bar{x}\bar{y}\bar{z})$.

Solution:



Minimization of Circuits

- Efficiency of a circuit: number of arrangement of its gates.
- Designing a circuit:
 - table specifying the output for each combination of input values.
 - May use sum-of-products expansion to find a set of logic gates that will implement this circuit.
- The expansion may contain unnecessary terms.

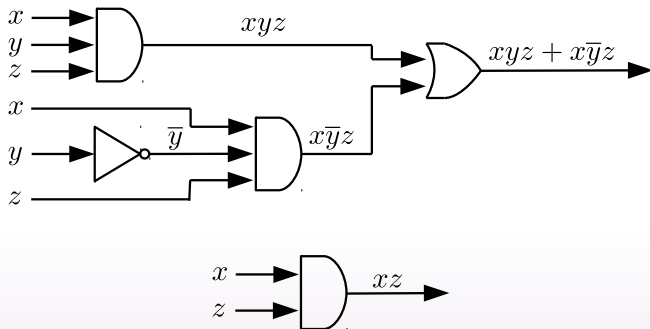
For instance, consider the circuit that has output 1 if and only if $x = y = z = 1$ or $x = z = 1$ and $y = 0$.

The sum-of-products expansion of this circuit is $xyz + x\bar{y}z$.

The two products in this expansion differ in exactly one variable, namely, y .

They can be combined as $xyz + x\bar{y}z = (y + \bar{y})(xz) = 1 \cdot (xz) = xz$.

Minimization of Circuits: Examples



xz is a Boolean expression with fewer operators.

The second circuit uses only one gate.

The first circuit uses three gates and an inverter.

Karnaugh Maps

- need to reduce the number of terms in a Boolean expression of a circuit
- it is necessary to find terms to combine.
- There is a graphical method, called a **Karnaugh map** or **K-map**, for finding terms to combine for Boolean functions involving a relatively small number of variables.

Karnaugh Maps: two variables

- Four minterms in sum-of-products expansion of a Boolean function in two variables x and y .
- A K-map consists of four cells, where a **1** is placed in the cell representing a minterm if this minterm is present in the expansion.
- Cells are said to be **adjacent** if the minterms that they represent differ in exactly one literal.
- For instance, the cell representing $\overline{x}y$ is adjacent to the cells representing xy and $\overline{x}\overline{y}$.

	y	$\overline{y}$
x	xy	$x\overline{y}$
$\overline{x}$	$\overline{x}y$	$\overline{x}\overline{y}$

Karnaugh Maps: two variables

Example: Find the K-maps for (a) $xy + \bar{x}y$, (b) $x\bar{y} + \bar{x}y$, and (c) $x\bar{y} + \bar{x}y + \bar{x}\bar{y}$.

Solution: We include a 1 in a cell when the minterm represented by this cell is present in the sum-of-products expansion.

	y	$\bar{y}$
x	1	
$\bar{x}$	1	

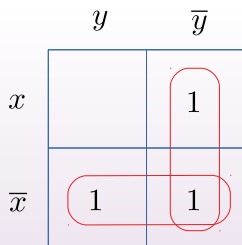
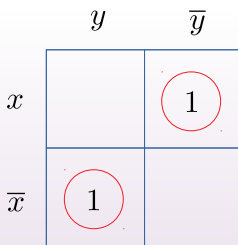
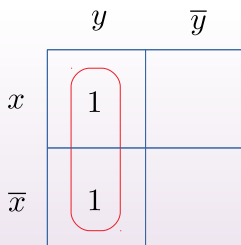
	y	$\bar{y}$
x		1
$\bar{x}$	1	

	y	$\bar{y}$
x		1
$\bar{x}$	1	1

Karnaugh Maps: two variables

Example: Simplify the sum-of-products expansions (a) $xy + \bar{x}y$, (b) $x\bar{y} + \bar{x}y$, and (c) $x\bar{y} + \bar{x}y + \bar{x}\bar{y}$.

Solution: The grouping of minterms is shown in the figures using the K-maps for these expansions. Minimal expansions for these sums-of-products are (a) y , (b) $\bar{x}y + x\bar{y}$, and (c) $\bar{x} + \bar{y}$.



Karnaugh Maps: three variables

Two cells are said to be **adjacent** if the minterms that they represent differ in exactly one literal.

One of the ways to form a K-map in three variables is:

	yz	$y\bar{z}$	$\bar{y}\bar{z}$	$\bar{y}z$
x	xyz	$xy\bar{z}$	$x\bar{y}\bar{z}$	$x\bar{y}z$
$\bar{x}$	$\bar{x}yz$	$\bar{x}y\bar{z}$	$\bar{x}\bar{y}\bar{z}$	$\bar{x}\bar{y}z$

Karnaugh Maps: three variables example

Example: Use K-maps to minimize these sum-of-products expansions.

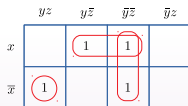
(a) $xy\bar{z} + x\bar{y}\bar{z} + \bar{x}yz + \bar{x}\bar{y}\bar{z}$

(b) $x\bar{y}z + x\bar{y}\bar{z} + \bar{x}yz + \bar{x}\bar{y}z + \bar{x}\bar{y}\bar{z}$

(c) $xyz + xy\bar{z} + x\bar{y}z + x\bar{y}\bar{z} + \bar{x}yz + \bar{x}\bar{y}z + \bar{x}\bar{y}\bar{z}$

(d) $xy\bar{z} + x\bar{y}\bar{z} + \bar{x}\bar{y}z + \bar{x}\bar{y}\bar{z}$

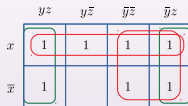
Solution:



(a) $x\bar{z} + \bar{y}\bar{z} + \bar{x}yz$



(b) $\bar{y} + \bar{x}z$



(c) $x + \bar{y} + z$



(d) $x\bar{z} + \bar{x}\bar{y}$

Reason: (d) The prime implicants $x\bar{z}$ and $\bar{x}\bar{y}$ are essential, but $\bar{y}\bar{z}$ is not essential, because its cells are covered by other two.